MODULE 1

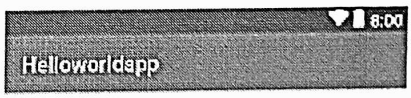
1. Develop a simple "Hello World" Android application demonstrating project creation, Android Studio interface, Android components, AVD creation, and application execution.

Activity_Main.xml file as follow→

```
<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

<TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:textSize="30dp"
      android:textColor="@color/colorPrimaryDark"
      android:background="@color/colorAccent"
      android:text="Hello World!"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintLeft_toLeftOf="parent"
      app:layout_constraintRight_toRightOf="parent"
      app:layout_constraintTop_toTopOf="parent" />

</android.support.constraint.ConstraintLayout>
```



MainActivity.java →

```
package com.anu.helloworldapp;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity
{
   @Override
   protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
}
}
```





2. Develop an Android application demonstrating the use of Programming Resources (colors.xml, themes.xml, strings.xml, dimens.xml, drawable resources) and apply them in UI components.

Color resource :

colors.xml as follow→

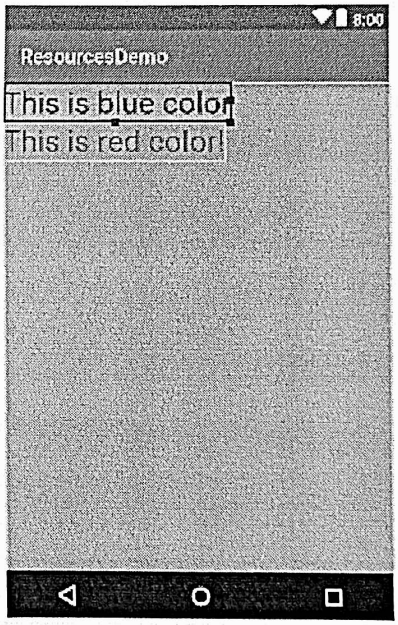
```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<color name="colorPrimary">#008577</color>
<color name="colorPrimaryDark">#00574B</color>
<color name="colorAccent">#D81B60</color>
<color name="red">#FF0000</color>
<color name="blue">#0000FF</color>
<color name="green">#00FF00</color>
</resources>
```

Activity_Main.xml→

```
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:background="@color/green"
   tools:context=".MainActivity">

 <TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:textSize="30dp"
      android:text="This is blue color"
      android:textColor="@color/blue"/>

<TextView
     android:layout_width="wrap_content"
     android:layout_height="wrap_content"
     android:text="This is red color!"
     android:textSize="30dp"
     android:textColor="@color/red"
     />
</LinearLayout>
```



Dimension Resource:

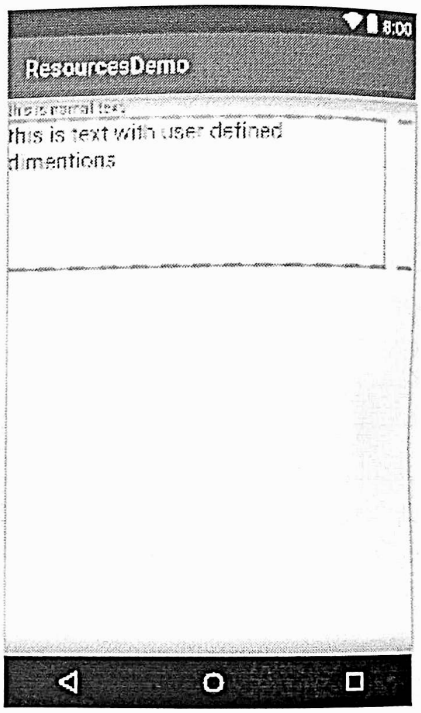
dimens.xml →

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<dimen name="height">35dp</dimen>
<dimen name="width">150dp</dimen>
<dimen name="size">20dp</dimen>
</resources>
```

Activity_Main.xml→

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
tools:context=".MainActivity">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="this is nornal text"/>

<TextView
android:layout_width="@dimen/width"
android:layout_height="@dimen/height"
android:textSize="@dimen/size"
android:text="this is text with user defined dimentions"/>
</LinearLayout>
```



Common Units Used

Unit	Meaning	Used For
dp	Density-independent pixels	Layout width, height, padding, margin
sp	Scale-independent pixels	Text size
px	Pixels	Rarely recommended
pt	Points	Printing-related measurements
mm	Millimeters	Physical measurement
in	Inches	Physical measurement

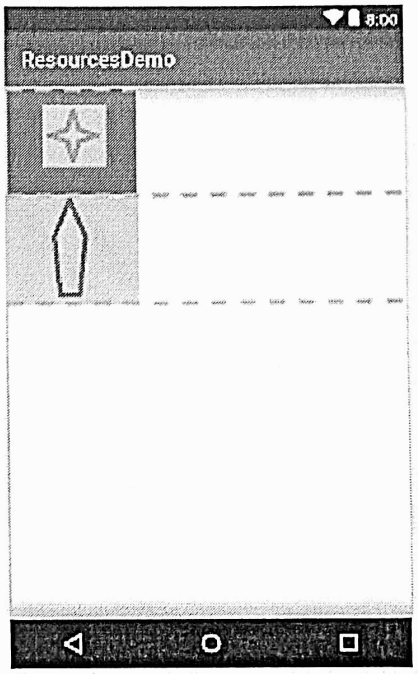
Drawable Resource:

1) First create minimum 2images in paint brush. Save at desired location.
2) Open file explorer ,go to the desired location, copy that images and paste the images by right click on drawable directory under res. [i.e. app→res→drawable] in android studio.

Activity_Main.xml→

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  tools:context=".MainActivity">

<ImageView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:src="@drawable/Untitled"/>
<ImageView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:src="@drawable/Untitled1"/>
</LinearLayout>
```



String Resources:

To open strings.xml file :

app→res→values→strings.xml

write code as follow in strings.xml →

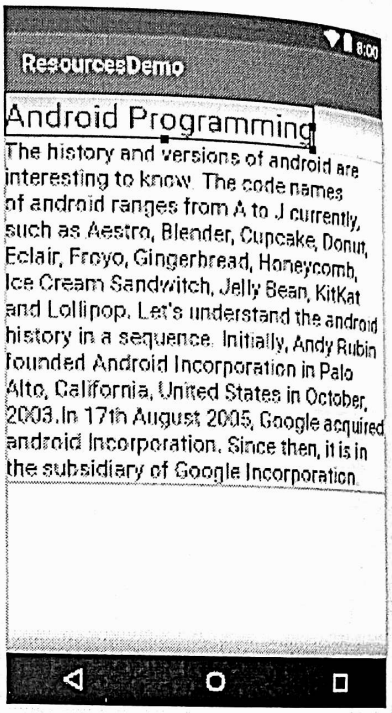
```
<resources>
<string name="app_name">ResourcesDemo</string>
<string name="Heading">Android Programming</string>
<string name="History">
 The history and versions of android are interesting to know. The code names of android ranges from A to J currently, such as Aestro, Blender, Cupcake, Donut, Eclair, Froyo, Gingerbread, Honeycomb, Ice Cream Sandwitch, Jelly Bean, KitKat and Lollipop.  Let's understand the android history in a sequence. Initially, Andy Rubin founded Android Incorporation in Palo Alto,California, United States in October, 2003.In 17th August 2005,
    Google acquired android Incorporation. Since then, it is in the subsidiary of Google Incorporation.
</string>
</resources>
```

Activity_Main.xml→

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   tools:context=".MainActivity">

<TextView
      android:id="@+id/textView"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:textSize="30dp"
      android:text="@string/Heading"
      android:textColor="@color/red"/>

 <TextView android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="@string/History"
       android:textSize="20dp"
       android:textColor="@color/blue"
        />
  </LinearLayout>
```



Theme resourse

Theme.xml→

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.Pract7" parent="Theme.Material3.DayNight">
    </style>

    <style name="Theme.Pract7" parent="Base.Theme.Pract7" />
</resources>
```

Note: if you change theme name then set that name in manifest file .

3. Develop an Android application demonstrating complete Activity Lifecycle methods and analyze execution using Logcat.

MainActivity.java →

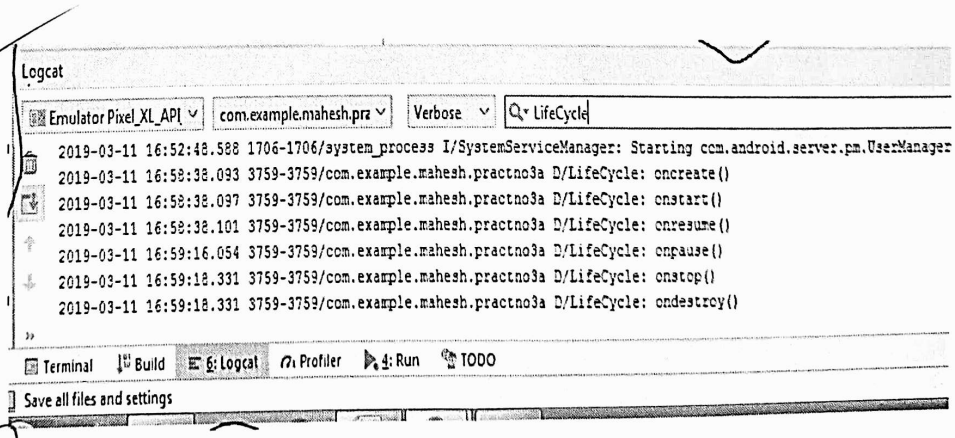
Note: To add following lifecycle methods , right click in mainactivity.java file and select Generate→override methods OR shortcut key is "CTRL+O".

```
package com.example.mahesh.practno3a;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.util.Log;

public class MainActivity extends AppCompatActivity
{
String tag="LifeCycle";

  @Override
  protected void onCreate(Bundle savedInstanceState)
  {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    Log.d(tag,"oncreate()");

  }
  public void onStart()
  {
    super.onStart();
    Log.d(tag,"onstart()");
  }
  public void onRestart()
  {
    super.onRestart();
    Log.d(tag,"onrestart()");
  }
  public void onResume()
  {
    super.onResume();
    Log.d(tag,"onresume()");
  }
  public void onPause()
  {
    super.onPause();
    Log.d(tag,"onpause()");
  }
  public void onStop()
  {
    super.onStop();
    Log.d(tag,"onstop()");
  }
  public void onDestroy()
  {
    super.onDestroy();
    Log.d(tag,"ondestroy()");
  }
}
```



4. Develop an Android application to create and integrate a Fragment and demonstrate Fragment lifecycle methods.

Create a new file as follow:

Right click app→new→Android Resource File→Name it as Lifeoffragment→click ok.

Lifeoffragment.java file as follow:-

```
package com.example.mahesh.practno3b;
import android.content.Context;
import android.os.Bundle;
import android.support.annotation.NonNull;
import android.support.annotation.Nullable;
import android.support.v4.app.Fragment;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.util.Log;

public class LifeOfFragment extends Fragment
{
   String tag="LifeCycle";

public LifeOfFragment()
 {
// Required empty public constructor
 }

    @Override
 public View onCreateView(LayoutInflater inflater,
 ViewGroup container,
                  Bundle savedInstanceState)
 {
 // Inflate the layout for this fragment
 return inflater.inflate(R.layout.fragment_life_of,
 container, false);
    }

     @Override
  public void onAttach(Context context)
  {
  super.onAttach(context);
     Log.d(tag,"on attach");
     }

     @Override
  public void onCreate(@Nullable Bundle
  savedInstanceState)
{
super.onCreate(savedInstanceState);
      Log.d(tag,"on create");
   }

    @Override
public void onViewCreated(@NonNull View view,
@Nullable Bundle savedInstanceState)
{
super.onViewCreated(view, savedInstanceState);
      Log.d(tag,"on viewcreate");
   }

    @Override
 public void onPause()
 {
 super.onPause();
      Log.d(tag,"on pause");
   }

    @Override
 public void onStop()
 {
 super.onStop();
      Log.d(tag,"on stop");
    }

    @Override
 public void onDestroyView()
 {
 super.onDestroyView();
      Log.d(tag,"on destroyview");
    }

    @Override
 public void onDestroy()
 {
 super.onDestroy();
      Log.d(tag,"on destory");
    }

    @Override
 public void onActivityCreated(@Nullable Bundle
 savedInstanceState)
 {
 super.onActivityCreated(savedInstanceState);
      Log.d(tag,"on act creation");
    }
```

```
    @Override
public void onStart()
{
super.onStart();
      Log.d(tag,"on start");
    }

    @Override
public void onResume()
{
super.onResume();
      Log.d(tag,"on resume");
    }
}
```

fragment_life_of.xml file as follow:-

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout

xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".LifeOfFragment">

<!-- TODO: Update blank fragment layout -->
<TextView
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:text="check me" />

</FrameLayout>
```

Activity_main.xml file as follow:-

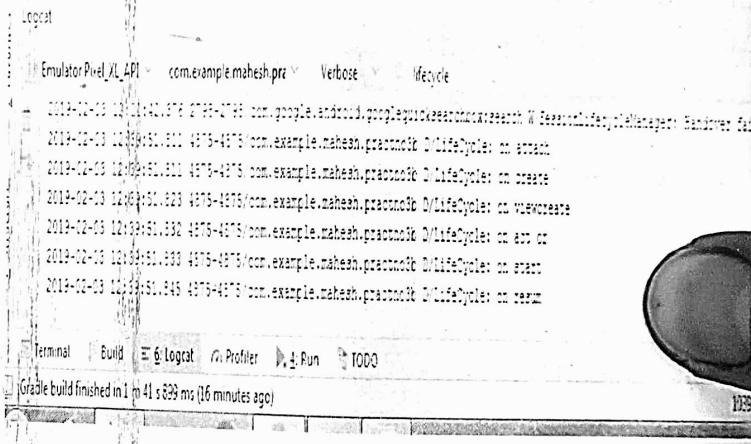
```
<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout

xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

<fragment
      android:id="@+id/test_fragment"

class="com.example.mahesh.practno3b.LifeOfFragment"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      tools:layout="@layout/fragment_life_of"
/>

</android.support.constraint.ConstraintLayout>
```



5. Develop an Android application implementing Fragment-based UI navigation between multiple screens.

Steps:

1) Create 2 fragments named FirstFragment and SecondFragment
2) Right clk on App→new→fragment→fragment(blank)

Parse data → send Data

Activity_main.xml→

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:id="@+id/fc"
  android:layout_width="match_parent"
  android:layout_height="match_parent"   > </FrameLayout>
```

fragment_first.xml→

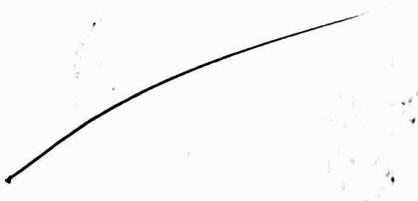
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical">

  <Button
    android:id="@+id/btnnext"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="go tosecond fragment"/>
</LinearLayout>
```

fragment_second.xml→

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical">

  <Button
    android:id="@+id/btnback"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="back to first fragment"/>
</LinearLayout>
```



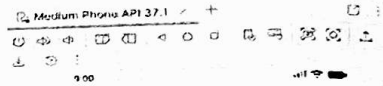
MainActivity.java→

```
package com.example.myapplication;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        if(savedInstanceState==null)
        {
            getSupportFragmentManager().beginTransaction().replace(R.id.fc,new FirstFragment()).commit();
        }
    }
}
```



firstfragment.java→

change the code of onCreateView() rest code as it is.

```
@Override
public View onCreateView(LayoutInflater inflater, ViewGroup container,
                         Bundle savedInstanceState)
{
    View v=inflater.inflate(R.layout.fragment_first,container,false);
    Button btnnext=v.findViewById(R.id.btnnext);
    btnnext.setOnClickListener(v1 -> {
getParentFragmentManager().beginTransaction().replace(R.id.fc,new
SecondFragment()).addToBackStack(null).commit();    } );
    return v;
}
```

argument

push





Secondfragment.java→

```
@Override
public View onCreateView(LayoutInflater inflater, ViewGroup container,
                         Bundle savedInstanceState)
{
    View v=inflater.inflate(R.layout.fragment_second,container,false);
    Button btnback=v.findViewById(R.id.btnback);
    btnback.setOnClickListener(v1 -> {
getParentFragmentManager().popBackStack();     } );
    return v;
}
```





6. Develop an Android application demonstrating communication between Activity and Fragment.

Theory

An **Activity** acts as the main entry point of an Android application, while a **Fragment** represents a reusable portion of the user interface.

Communication between an Activity and a Fragment can occur in two ways:

1. **Activity → Fragment:** Data is passed using a Bundle.
2. **Fragment → Activity:** Data is sent back using an interface callback implemented by the Activity.

This approach promotes modular, reusable, and maintainable Android applications.

Add blank fragment as added in the last practical.

Activity_main.xml→

```
?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:background="@color/cyan"
  android:orientation="vertical"
  android:padding="16dp">

  <TextView
    android:id="@+id/txtActivity"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Message gets display here once received from fragment"
    android:textSize="20sp"
    android:padding="10dp"/>

  <FrameLayout
    android:id="@+id/fragment_container"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
</LinearLayout>
```

I. ainactivity.java→

```
package com.example.pract6;
import
androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.TextView;

public class MainActivity extends
AppCompatActivity
    implements
MessageFragment.OnMessageListener {

  TextView txtActivity;

  @Override
  protected void onCreate(Bundle
savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    txtActivity = findViewById(R.id.txtActivity);

    // Create Fragment
    MessageFragment fragment = new
MessageFragment();

    // Pass data to Fragment
    Bundle bundle = new Bundle();
    bundle.putString("message", "Hello
Fragment!");
    fragment.setArguments(bundle);

    getSupportFragmentManager()
        .beginTransaction()
        .replace(R.id.fragment_container,
fragment)
        .commit();
  }

  // Receive message from Fragment
  @Override
  public void onMessageReceived(String
message) {
    txtActivity.setText(message);
  }
}
```

messagefragment.xml→

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:background="@color/light"
  android:gravity="center"
  android:orientation="vertical">
```

```
    <TextView
        android:id="@+id/txtFragment"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="22sp"/>

    <Button
        android:id="@+id/btnSend"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Message to Activity"/>

</LinearLayout>
```

messageFragment.java→

```
package com.example.pract6;

import android.content.Context;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;

public class MessageFragment extends
Fragment {

    TextView txtFragment;
    Button btnSend;

    OnMessageListener listener;

    // Interface
    public interface OnMessageListener {
        void onMessageReceived(String message);
    }

    @Override
    public void onAttach(@NonNull Context
context) {
        super.onAttach(context);

        if (context instanceof OnMessageListener) {
            listener = (OnMessageListener) context;
        }
    }

    @Override
    public View onCreateView(LayoutInflater
inflater,
                             ViewGroup container,
                             Bundle savedInstanceState) {

        View view = inflater.inflate(
                R.layout.fragment_message,
                container,
                false);

        txtFragment =
view.findViewById(R.id.txtFragment);
        btnSend =
view.findViewById(R.id.btnSend);

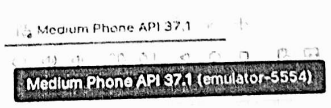
        // Receive data from Activity
        Bundle bundle = getArguments();

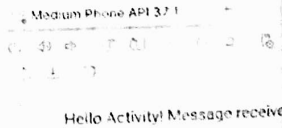
        if (bundle != null) {

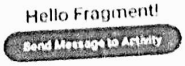
txtFragment.setText(bundle.getString("message
"));
        }

        // Send data back to Activity
        btnSend.setOnClickListener(v -> {
            listener.onMessageReceived(
                    "Hello Activity! Message received
from Fragment.");
        });

        return view;
    }
}
```









7. Develop an Android application using LinearLayout (Horizontal and Vertical) to design structured UI.

Mainactivity.xml→

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp">

    <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Student Registration"
      android:textSize="24sp"
      android:textStyle="bold"
      android:layout_gravity="center"
      android:padding="10dp"/>

    <LinearLayout
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:orientation="horizontal"
      android:padding="5dp">

      <TextView
        android:layout_width="100dp"
        android:layout_height="wrap_content"
        android:text="Name"/>

      <EditText
        android:id="@+id/etName"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:hint="Enter Name"/>

    </LinearLayout>

    <LinearLayout
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:orientation="horizontal"
      android:padding="5dp">

      <TextView
        android:layout_width="100dp"
        android:layout_height="wrap_content"
        android:text="Email"/>

      <EditText
        android:id="@+id/etEmail"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:hint="Enter Email"/>

    </LinearLayout>

    <LinearLayout
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:orientation="horizontal"
      android:padding="5dp">

      <TextView
        android:layout_width="100dp"
        android:layout_height="wrap_content"
        android:text="Phone"/>

      <EditText
        android:id="@+id/etPhone"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:hint="Enter Phone"
        android:inputType="phone"/>

    </LinearLayout>

      <LinearLayout
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:orientation="horizontal"
      android:gravity="center"
      android:paddingTop="20dp">

      <Button
        android:id="@+id/btnSubmit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Submit"/>
```

```
    <Button
      android:id="@+id/btnClear"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Clear"
      android:layout_marginStart="20dp"/>
  </LinearLayout>
</LinearLayout>
```

Mainactivity.java→

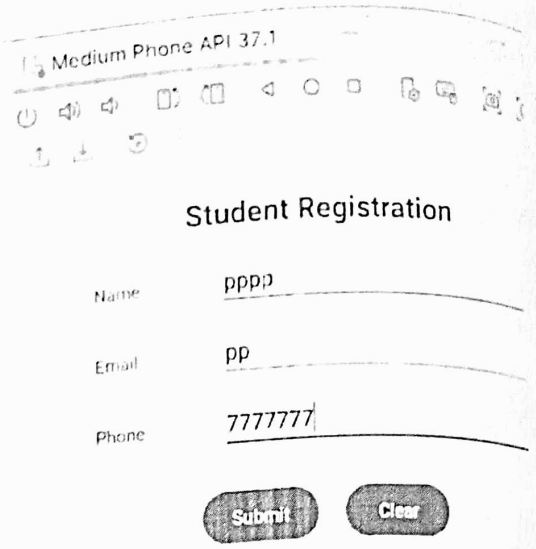
```
package com.example.pract7;
import
androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends
AppCompatActivity {
  EditText etName, etEmail, etPhone;
  Button btnSubmit, btnClear;
  @Override
  protected void onCreate(Bundle
savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    etName = findViewById(R.id.etName);
    etEmail = findViewById(R.id.etEmail);
    etPhone = findViewById(R.id.etPhone);
    btnSubmit = findViewById(R.id.btnSubmit);
    btnClear = findViewById(R.id.btnClear);
    btnSubmit.setOnClickListener(v -> {
      String name =
etName.getText().toString();
      Toast.makeText(this, "Welcome " +
name,  Toast.LENGTH_SHORT).show();

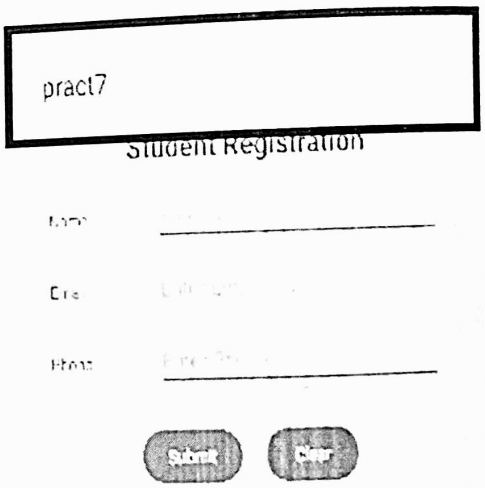
    });

    btnClear.setOnClickListener(v -> {
      etName.setText("");
      etEmail.setText("");
      etPhone.setText("");

    });
  }
}
```



Note: A toast message appears when the submit button is clicked, and data typed in text fields is cleared when the clear button is clicked.



8. Develop an Android application using RelativeLayout to align UI components dynamically.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:padding="20dp"
  tools:context=".MainActivity">

  <TextView
    android:id="@+id/txtTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Student Registration"
    android:textSize="24sp"
    android:textStyle="bold"
    android:layout_centerHorizontal="true"/>

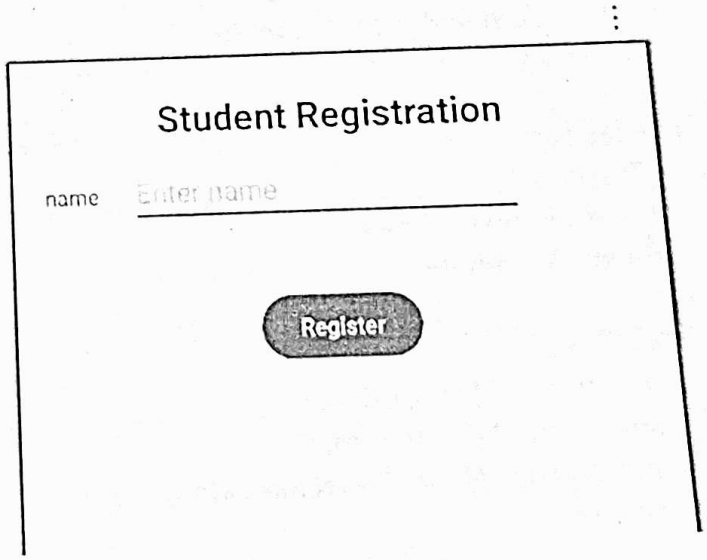
  <TextView
    android:id="@+id/txtname"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="name"

    android:layout_marginTop="55dp"/>

  <EditText
    android:id="@+id/edtname"
    android:layout_width="250dp"
    android:layout_height="wrap_content"
    android:layout_toEndOf="@id/txtname"
    android:layout_alignBaseline="@id/txtname"
    android:layout_marginStart="20dp"
    android:hint="Enter name"/>

    <Button
    android:id="@+id/btnRegister"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Register"
    android:layout_below="@id/edtname"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="40dp"/>

</RelativeLayout>
```



9. Develop an Android application using TableLayout to design a structured Login/Registration Form.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TableLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:padding="20dp"
  android:stretchColumns="1">

  <TableRow>
    <TextView
      android:layout_span="2"
      android:gravity="center"
      android:text="Student Registration"
      android:textSize="24sp"
      android:textStyle="bold"
      android:paddingBottom="20dp"/>
  </TableRow>

    <TableRow>
    <TextView
      android:text="Name"
      android:padding="8dp"/>

    <EditText
      android:id="@+id/etName"
      android:hint="Enter Name"/>
  </TableRow>

  <TableRow>
    <TextView
      android:text="Email"
      android:padding="8dp"/>

    <EditText
      android:id="@+id/etEmail"
      android:hint="Enter Email"
      android:inputType="textEmailAddress"/>
  </TableRow>

  <TableRow>
    <TextView
      android:text="Password"
      android:padding="8dp"/>

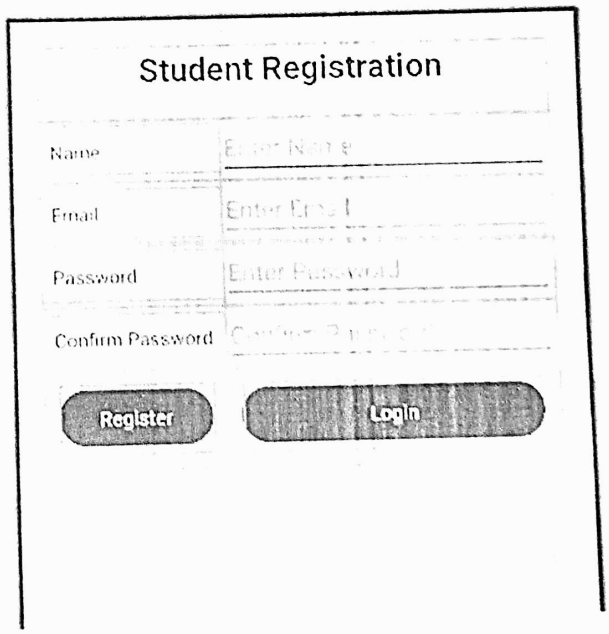
    <EditText
      android:id="@+id/etPassword"
      android:hint="Enter Password"
      android:inputType="textPassword"/>
  </TableRow>

  <TableRow
    android:gravity="center">

    <Button
      android:id="@+id/btnRegister"
      android:text="Register"
      android:layout_margin="10dp"/>

    <Button
      android:id="@+id/btnLogin"
      android:text="Login"
      android:layout_margin="10dp"/>
  </TableRow>

</TableLayout>
```



10. Develop an Android application using ScrollView to display long scrollable content.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="16dp">

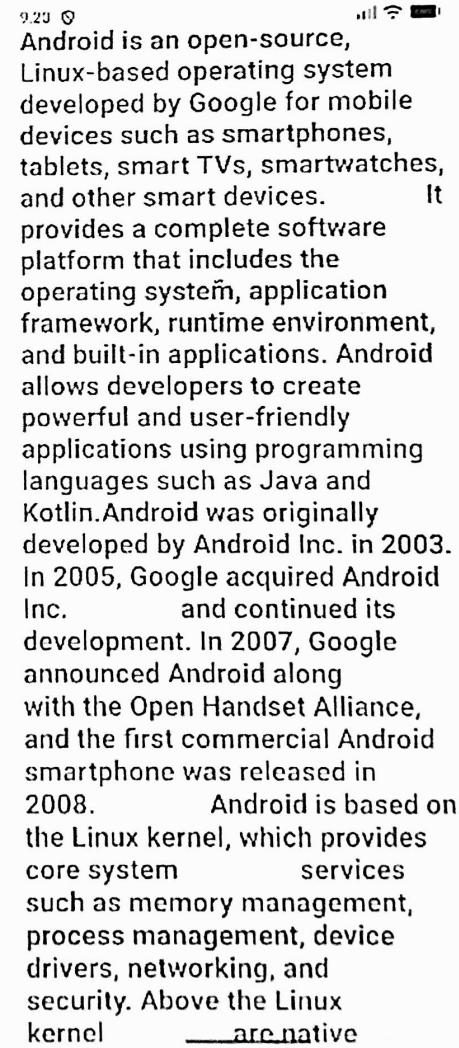
    <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Android is an open-source,
      Linux-based operating system developed by Google for mobile devices such as smartphones, tablets,
smart TVs, smartwatches, and other smart devices. It provides a complete software platform that includes the
operating system, application framework, runtime environment, and built-in applications. Android allows
developers to create powerful and user-friendly applications using programming languages such as Java and
Kotlin.Android was originally developed by Android Inc. in 2003. In 2005, Google acquired Android Inc. and
continued its development. In 2007, Google announced Android
along with the Open Handset Alliance, and the first commercial
Android smartphone was released in 2008. Android is based on the
Linux kernel, which provides core system services such as memory
management, process management, device drivers, networking,
and security. Above the Linux kernel are native libraries and the
Android Runtime (ART), which execute Android applications
efficiently. The Application Framework provides APIs that
developers use to build applications, while the top layer consists
of system applications and user-installed applications.
Android applications are built using different application
components. An Activity represents a single screen with a user
interface. A Service performs long-running operations in the
background without direct user interaction. A Broadcast Receiver
responds to system-wide broadcast messages, such as battery
status or network changes. A Content Provider enables data
sharing between different applications. Android applications are
designed using XML layout files and Java or Kotlin code. The user
interface is created using layouts such as LinearLayout,
RelativeLayout, ConstraintLayout, TableLayout, FrameLayout, and
ScrollView. UI components such as TextView, EditText, Button,
ImageView, CheckBox, RadioButton, and Spinner are placed inside
these layouts to create interactive screens."

      android:textStyle="bold"
      android:textSize="24sp"
      android:paddingBottom="20dp"/>

  </LinearLayout>

</ScrollView>
```





11. Develop an Android application using AbsoluteLayout demonstrating fixed positioning.

Activity_main.xml

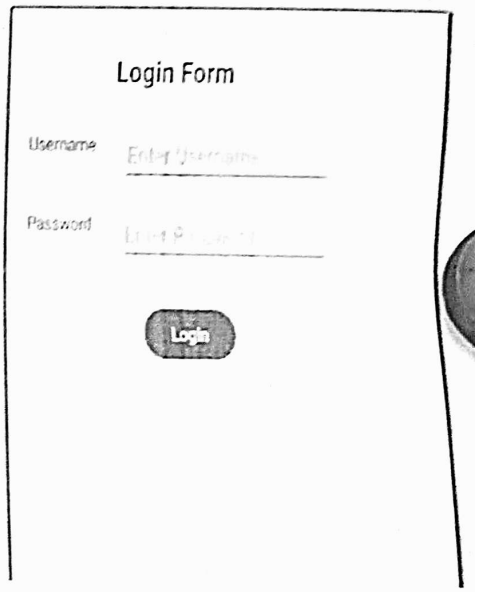
```
<?xml version="1.0" encoding="utf-8"?>

<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent">

  <TextView
    android:id="@+id/txtTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login Form"
    android:textSize="22sp"
    android:textStyle="bold"
    android:layout_x="100dp"
    android:layout_y="30dp"/>

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Username"
    android:layout_x="20dp"
    android:layout_y="90dp"/>

  <EditText
    android:id="@+id/etUsername"
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:hint="Enter Username"
    android:layout_x="110dp"
    android:layout_y="85dp"/>
 <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Password"
    android:layout_x="20dp"
    android:layout_y="150dp"/>
 <EditText
    android:id="@+id/etPassword"
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:hint="Enter Password"
    android:inputType="textPassword"
    android:layout_x="110dp"
    android:layout_y="145dp"/>
 <Button
   android:id="@+id/btnLogin"
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:text="Login"
   android:layout_x="130dp"
   android:layout_y="220dp"/>
</AbsoluteLayout>
```



12. Develop an Android application using ListView with ArrayAdapter and implement item click handling.

Activity_Main.xml →

```
<?xml version="1.0" encoding="utf-8"?>
<android.support.constraint.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

<ListView
android:id="@+id/listview1"
android:layout_width="wrap_content "
android:layout_height="wrap_content">
</ListView>
</android.support.constraint.ConstraintLayout>
```

MainActivity.java →

```
package com.anu.listviewdemo;

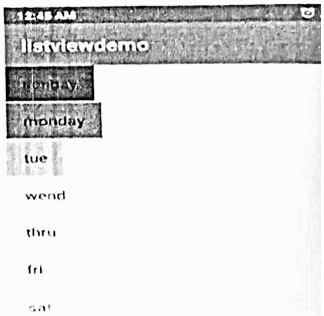
import android.graphics.Color;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.TextView;
import android.widget.Toast;
```

```
public class MainActivity extends AppCompatActivity implements  AdapterView.OnItemClickListener {
ListView l;
   String[] days={"sunday","monday","tue","wend","thru","fri","sat"};

   @Override
   protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
      l=(ListView) findViewById(R.id.listview1);
ArrayAdapter<String> adapter=new
ArrayAdapter<String>(this,android.R.layout.simple_list_item_1,days);
l.setAdapter(adapter);
l.setOnItemClickListener(this);

   }

   @Override
   public void onItemClick(AdapterView<?> parent, View view, int position, long id)
   {
TextView temp=(TextView) view;
Toast.makeText(this,temp.getText()+"  "+position,Toast.LENGTH_SHORT).show();
      if(position ==0)
      {
temp.setBackgroundColor(Color.BLUE);
      }
      if(position ==1)
      {
temp.setBackgroundColor(Color.RED);
      }
      if(position ==2)
      {
temp.setBackgroundColor(Color.CYAN);
      }
   }
}
```



OUTPUT: onclick of list item the background color of that list item row, is getting changed.

Note: As you can see in the above code events are created only for 1st three list items. If you want you can have for all list items mentioned in array "days".

13. Develop an Android application implementing AppBar (Toolbar) and Options Menu.

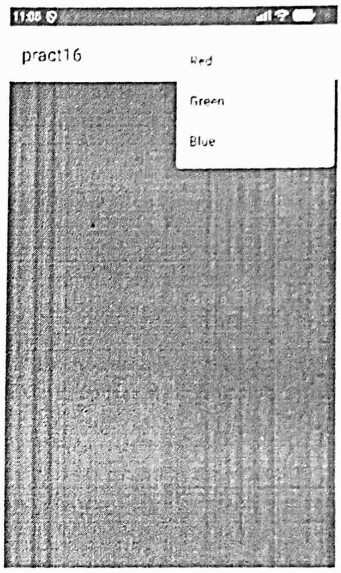
Note:

1) **No change in activity XML file.**
2) **Change manifest.xml for the mentioned line only :** res → values → themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <style name="Base.Theme.Pract16" parent="Theme.Material3.DayNight">

    </style>

    <style name="Theme.Pract16" parent="Base.Theme.Pract16" />
</resources>
```



Create mymenu.xml file as follow:

Right click on res→new→android **new resource file**

name:mymenu

resource type:menu

root elemtn:menu

mymenu.xml→

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/r1"
        android:title="Red"/>

    <item
        android:id="@+id/g1"
        android:title="Green"/>

    <item
        android:id="@+id/b1"
        android:title="Blue"/>

</menu>
```

Java file →

```
package com.example.pract16;

import android.graphics.Color;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.LinearLayout;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  LinearLayout layout;

  @Override
  protected void onCreate(Bundle savedInstanceState)
  {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    layout = findViewById(R.id.layout);
  }

  @Override
  public boolean onCreateOptionsMenu(Menu menu)
  {

    getMenuInflater().inflate(R.menu.mymenu, menu);
    return true;
  }

  @Override
  public boolean onOptionsItemSelected(MenuItem item) {
    int id = item.getItemId();

    if (id == R.id.r1) {
      layout.setBackgroundColor(Color.RED);
    } else if (id == R.id.g1) {
      layout.setBackgroundColor(Color.GREEN);
    } else if (id == R.id.b1) {
      layout.setBackgroundColor(Color.BLUE);

    }
  return true;
  }
}
```

Practical practical no: 14

3. Develop an Android application implementing contextual menus and popup menus.

activitymain.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">

   <TextView
      android:id="@+id/txtMessage"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Long Press Me"
      android:textSize="22sp"
      android:padding="20dp"/>

   <Button
      android:id="@+id/btnPopup"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Show Popup Menu"
      android:layout_marginTop="30dp"/>

</LinearLayout>
```

Create 2 menu files named as context_menu and popup_menu

Right click on res→new→new resource file

name:mymenu

resource type:menu

root elemtn:menu

context_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
   <item
      android:id="@+id/r1"
      android:title="red"/>
```

```
  <item
     android:id="@+id/g1"
     android:title="green"/>
</menu>
```

popup_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
   <item
      android:id="@+id/h1"
      android:title="Singing"/>

   <item
      android:id="@+id/h2"
      android:title="Dancing"/>
</menu>
```

Mainactivity.java

```
package com.example.pract18;

import android.os.Bundle;
import android.view.*;
import android.view.View;
import android.widget.*;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
   TextView txtMessage;
   Button btnPopup;
   @Override
   protected void onCreate(Bundle savedInstanceState)
   {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      txtMessage = findViewById(R.id.txtMessage);
      btnPopup = findViewById(R.id.btnPopup);
      // Register Context Menu
      registerForContextMenu(txtMessage);

      // Popup Menu
      btnPopup.setOnClickListener(v -> {
         PopupMenu popupMenu = new PopupMenu(MainActivity.this, btnPopup);
         popupMenu.getMenuInflater().inflate(R.menu.popup_menu, popupMenu.getMenu());
         popupMenu.setOnMenuItemClickListener(item -> {
            if (item.getItemId() == R.id.h1)
            {
```

```
            Toast.makeText(MainActivity.this,"my hobby is singing",Toast.LENGTH_SHORT).show();
        }
        else if (item.getItemId() == R.id.h2)
        {
            Toast.makeText(MainActivity.this,"my hobby is dancing",Toast.LENGTH_SHORT).show();
        }
        return true;
      });
      popupMenu.show();
    });
  }

  @Override
  public void onCreateContextMenu(ContextMenu menu,View v,ContextMenu.ContextMenuInfo
menuInfo)
  {
    super.onCreateContextMenu(menu, v, menuInfo);
    MenuInflater inflater = getMenuInflater();
    inflater.inflate(R.menu.context_menu, menu);
  }

  @Override
  public boolean onContextItemSelected(@NonNull MenuItem item) {

    if (item.getItemId() == R.id.r1) {
      Toast.makeText(this,"red color selected",Toast.LENGTH_SHORT).show();
      return true;

    }
    else if (item.getItemId() == R.id.g1) {
      Toast.makeText(this,"green color selected",Toast.LENGTH_SHORT).show();
      return true;
    }

    return super.onContextItemSelected(item);
  }
}
```

pract18

pract18

pract18

Long Press Me

Show Popup Menu

Long Press Me

red

green

Long Press Me

Singing

Dancing

14.2

15. Develop an android app that performs basic Arithmetic operations.

Activity_main.xml→

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp">

<EditText
android:id="@+id/num1"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter first number"
android:inputType="numberDecimal" />

<EditText
android:id="@+id/num2"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter second number"
android:inputType="numberDecimal" />

<Button
android:id="@+id/add"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="ADD" />

<Button
android:id="@+id/sub"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="SUBTRACT" />

<Button
android:id="@+id/mul"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="MULTIPLY" />

<Button
android:id="@+id/div"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="DIVIDE" />

<TextView
android:id="@+id/result"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Result" android:textSize="22sp"
android:padding="20dp" />

</LinearLayout>
```

Mainactivity.java→

```
package com.example.arithmetic;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity
{

  EditText num1, num2;
  TextView result;
  Button add, sub, mul, div;

  @Override
  protected void onCreate(Bundle
savedInstanceState)
{
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    num1 = findViewById(R.id.num1);
    num2 = findViewById(R.id.num2);
    result = findViewById(R.id.result);
```

```
                result.setText("Result = " + ans);
            }
        }

        add = findViewById(R.id.add);
        sub = findViewById(R.id.sub);
        mul = findViewById(R.id.mul);
        div = findViewById(R.id.div);

        add.setOnClickListener(v -> calculate('+'));
        sub.setOnClickListener(v -> calculate('-'));
        mul.setOnClickListener(v -> calculate('*'));
        div.setOnClickListener(v -> calculate('/'));
    }

    void calculate(char operator)
    {

        double a =
Double.parseDouble(num1.getText().toString());
        double b =
Double.parseDouble(num2.getText().toString());

        double ans = 0;

        switch (operator) {
            case '+':
                ans = a + b;
                break;

            case '-':
                ans = a - b;
                break;

            case '*':
                ans = a * b;
                break;

            case '/':
                if (b != 0)
                    ans = a / b;
                else {
                    result.setText("Cannot divide by
zero");
                    return;
                }
                break;
        }
```

16.Develop an Android application implementing Explicit and Implicit Intents.

Activitymain.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <Button
        android:id="@+id/btnExplicit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Open Second Activity"/>

    <Button
        android:id="@+id/btnImplicit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Open Website"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

Create SecondActivity (right clk on app→new→activity→empty views activity→ name it → ok)

Secondactivity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Second Activity"
        android:textSize="24sp"/>

</LinearLayout>
```

Secondactivity.java

No change

Mainactivity.java

```
package com.example.pract19;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
  Button btnExplicit, btnImplicit;

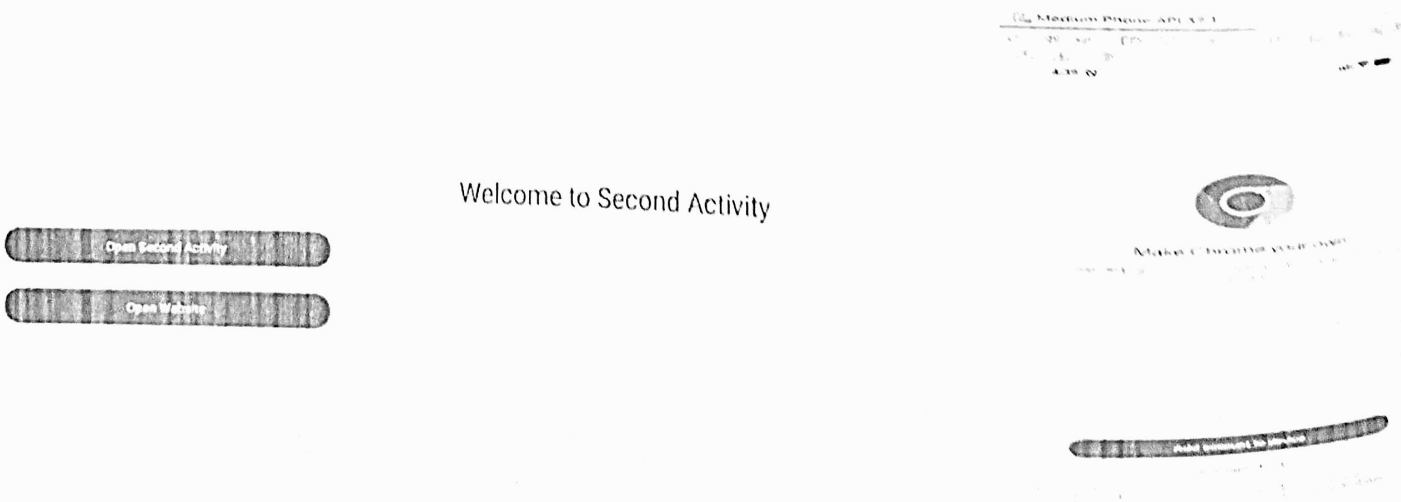
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    btnExplicit = findViewById(R.id.btnExplicit);
    btnImplicit = findViewById(R.id.btnImplicit);

    // Explicit Intent
    btnExplicit.setOnClickListener(v -> {
      Intent intent = new Intent(MainActivity.this,  SecondActivity.class);
      startActivity(intent);
    });

    // Implicit Intent
    btnImplicit.setOnClickListener(v -> {
      Intent intent = new Intent(Intent.ACTION_VIEW,  Uri.parse("https://www.google.com"));
      startActivity(intent);
    });
  }
}
```

output: on click of open second activity button second activity is getting open . when we click on open website button google chrome is getting open.



17. Develop an Android application implementing SQLite database CRUD operations.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/andr
oid"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp">

  <EditText
    android:id="@+id/etId"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter ID"/>

  <EditText
    android:id="@+id/etName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Name"/>

  <Button
    android:id="@+id/btnInsert"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Insert"/>

  <Button
    android:id="@+id/btnView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="View"/>

  <Button
    android:id="@+id/btnUpdate"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Update"/>

  <Button
    android:id="@+id/btnDelete"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Delete"/>

  <TextView
    android:id="@+id/txtResult"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="18sp"
    android:paddingTop="20dp"/>

</LinearLayout>
```

Create DBHelper java class

```
package com.example.pract25;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

  public DBHelper(Context context) {
    super(context, "StudentDB", null, 1);
  }

  @Override
  public void onCreate(SQLiteDatabase db) {
    db.execSQL("CREATE TABLE Student(id INTEGER
PRIMARY KEY,name TEXT)");
  }

  @Override
  public void onUpgrade(SQLiteDatabase db, int
oldVersion, int newVersion) {
    db.execSQL("DROP TABLE IF EXISTS Student");
    onCreate(db);
  }

  public boolean insertData(int id, String name) {
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues cv = new ContentValues();
    cv.put("id", id);
    cv.put("name", name);
    long result = db.insert("Student", null, cv);
    return result != -1;
  }
```

```
    public Cursor getData() {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.rawQuery("SELECT * FROM Student", null);
    }

    public boolean updateData(int id, String name) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues cv = new ContentValues();
        cv.put("name", name);
        int rows = db.update("Student", cv, "id=?", new
String[]{String.valueOf(id)});
        return rows > 0;
    }

    public boolean deleteData(int id) {
        SQLiteDatabase db = this.getWritableDatabase();
        int rows = db.delete("Student", "id=?", new
String[]{String.valueOf(id)});
        return rows > 0;
    }
}
```

mainactivity.java

```
package com.example.pract25;


import androidx.appcompat.app.AppCompatActivity;
import android.database.Cursor;
import android.os.Bundle;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    EditText etId, etName;
    Button btnInsert, btnView, btnUpdate, btnDelete;
    TextView txtResult;

    DBHelper db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etId = findViewById(R.id.etId);
        etName = findViewById(R.id.etName);

        btnInsert = findViewById(R.id.btnInsert);
        btnView = findViewById(R.id.btnView);
        btnUpdate = findViewById(R.id.btnUpdate);
        btnDelete = findViewById(R.id.btnDelete);

        txtResult = findViewById(R.id.txtResult);

        db = new DBHelper(this);

        btnInsert.setOnClickListener(v -> {

            int id = Integer.parseInt(etId.getText().toString());
            String name = etName.getText().toString();

            if(db.insertData(id,name))

Toast.makeText(this,"Inserted",Toast.LENGTH_LONG).sho
w();
            else

Toast.makeText(this,"Failed",Toast.LENGTH_LONG).show
;
        });

        btnView.setOnClickListener(v -> {
      Cursor c = db.getData();
            StringBuilder sb = new StringBuilder();
            while(c.moveToNext()){
                sb.append("ID :
").append(c.getInt(0)).append("\n");
                sb.append("Name :
").append(c.getString(1)).append("\n\n");
        }
            txtResult.setText(sb.toString());
    });

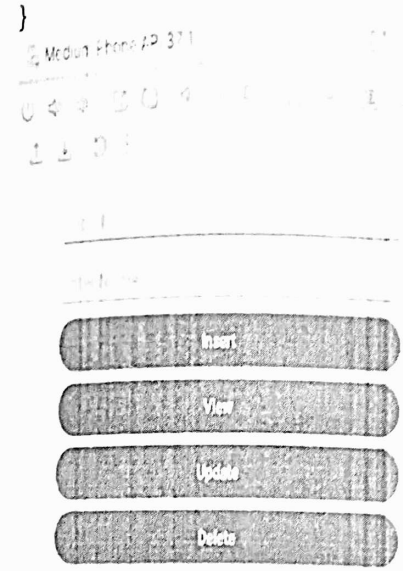
        btnUpdate.setOnClickListener(v -> {
            int id=Integer.parseInt(etId.getText().toString());
            String name=etName.getText().toString();
            if(db.updateData(id,name))

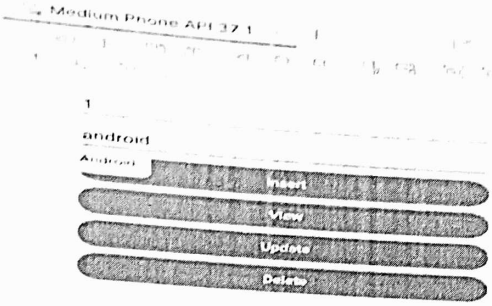
Toast.makeText(this,"Updated",Toast.LENGTH_LONG).sh
w();
            else
                Toast.makeText(this,"Not
Found",Toast.LENGTH_LONG).show();       });

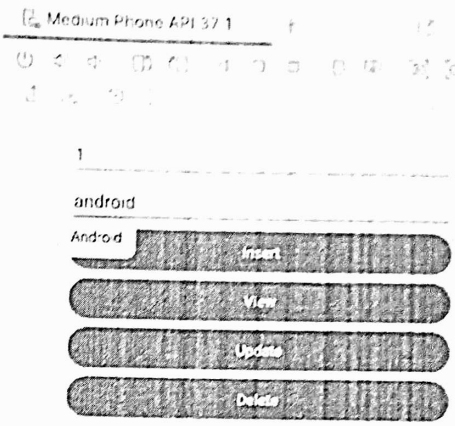
        btnDelete.setOnClickListener(v -> {
            int id=Integer.parseInt(etId.getText().toString());
            if(db.deleteData(id))

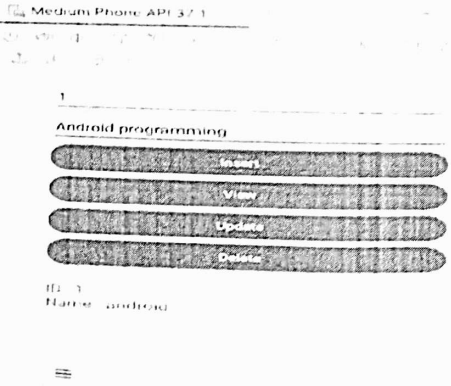
Toast.makeText(this,"Deleted",Toast.LENGTH_LONG).sho
();
```

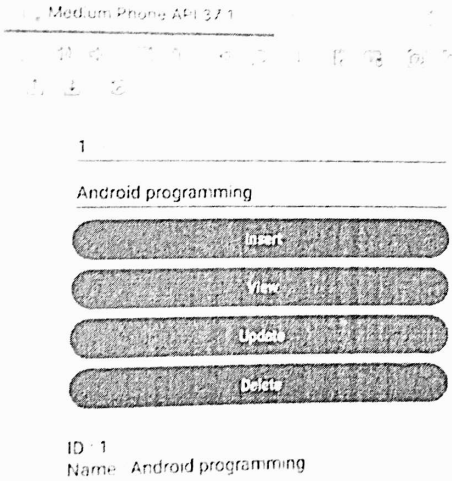
```
        else
            Toast.makeText(this,"Record Not
Found",Toast.LENGTH_LONG).show();
        });
    }
}
```

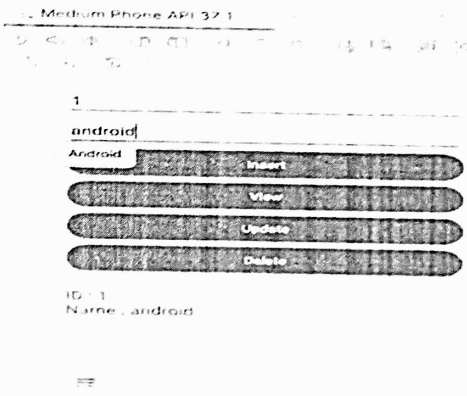












18. Activity to activity data sharing

Note : add one new empty vies activity name it as SecondAcivity ("Right clk on APP→new→activity→empty views activity→name it SecondActivity → click on finish)

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="20dp">

  <EditText
    android:id="@+id/name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter your name" />

  <Button
    android:id="@+id/btnSend"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Send Data" />
</LinearLayout>
```

MainActivity.java — First Activity

```
package com.example.myapp;
import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity
{
  EditText name;
  Button btnSend;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    name = findViewById(R.id.name);
    btnSend = findViewById(R.id.btnSend);
    btnSend.setOnClickListener(v -> {

      String data = name.getText().toString();
      Intent intent = new Intent(MainActivity.this,
SecondActivity.class);
      intent.putExtra("username", data);
      startActivity(intent);
    });
  }
}
```

Second activity→

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center">

  <TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textSize="24sp" />
</LinearLayout>
```

SecondActivity.java — Second Activity

```
package com.example.myapp;
import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;
public class SecondActivity extends AppCompatActivity {
```

```
TextView textView;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_second);
    textView = findViewById(R.id.textView);
    String data = getIntent().getStringExtra("username");
    textView.setText("Welcome " + data);
}
```



